

sustained outperformance from the value quintile following this latest brief period of underperformance.

The update to Lakonishok et al.'s research demonstrates that, aside from short periods of underperformance, value stocks generate a consistent value premium, and beat both the market and glamour stocks over the long haul. The reasons why are a source of some controversy. Behavioral finance economists like De Bondt and Thaler and Lakonishok, Shleifer, and Vishny believe that they do so because they are contrarian to overreaction and naïve extrapolation. The alternative explanation for the superior returns to value strategies, argued by influential efficient market economists Eugene Fama and Ken French, is that they are *fundamentally riskier*. Value strategies seem to outperform, claim Fama and French, as compensation for this additional risk. The same argument was used to criticize De Bondt and Thaler's research into overreaction. To be fundamentally riskier, value stocks must underperform glamour stocks with some frequency, and particularly in what Lakonishok et al. describe as "bad" states of the world, such as extreme down markets or recessions. Lakonishok et al. found that while the value strategy does disproportionately well in good times, its performance in bad times is also impressive. In fact, over longer horizons value strategies almost always outperform glamour strategies, and they do particularly well in those bad states of the world. They could not find any measure of risk that explained the huge difference in average returns to the value strategy. They were able to find, however, evidence that investors appear to be extrapolating past performance, even though the future does not warrant such extrapolation.

Lakonishok et al. describe the essence of naïve extrapolation as the idea that investors are excessively optimistic about glamour stocks because they tie their expectations of future growth to past growth, and excessively pessimistic about value stocks for the same reason. They tested the validity of this assertion by comparing the actual growth rates of fundamental measures like sales, earnings, and operating income to past and expected rates of growth of the same fundamental measures. Lakonishok et al. calculated expected growth rates from each security's price ratios. If the market puts a high price on a security relative to its fundamentals, it is implicitly expecting high growth in those fundamentals. In other words, for example, the higher the price-to-earnings ratio, the faster the market expects earnings to grow. We can show this with Gordon's constant growth model, which is a model for determining the intrinsic value of a stock given a series of dividends that grow at a constant rate.²⁴ The equation is as follows:

$$P = \frac{D_1}{r - g}$$